# PHYSICAL CHEMISTRY CUMULATIVE EXAM

**December 2, 1995**

0.1 mole of He, considered as an ideal gas is contained in 1 L volume at 0°C . The 1 L container is connected, through an initially closed valve, to a larger, evacuated vessel of 2 L volume. The whole setup is kept in an isothermal calorimeter. The valve is suddenly opened.

(A) After equilibrium has been reached, how much heat has been absorbed by the system from the calorimeter (in Joules)?

(B) How much is the entropy change in J/K ? ($R=8.3145\ JK^{-1}mol^{-1}$).

(C) Helium is a nearly ideal gas but carbon dioxide is not. If 0.1 mol He is exchanged for 0.1 mol $CO_2$ is question (A), what is the answer (how much heat has been absorbed from the caorimeter, in J units)? The isothermal Joule-Thompson coefficient of $CO_2$, i.e. the derivative of the molar enthalpy with respect to pressure at constant temperature, $(\partial H/\partial p)_T=-41.2\ J\ atm^{-1}\ mol^{-1}$. For the calculations of the pressures, you can still use the ideal gas laws, as no great accuracy is required. $R=0.08206\ L\ atm\ K^{-1}mol^{-1}$.

(D) Suppose that the 1 L vessel still contains He but the 2L one is contains neon at the same pressure as the He. How much does the entropy change after the valve is opened and the gases are mixed? Consider both He and Ne ideal gases.